

Investigating the Efficiency-Enforcing Role of Biologically-Inspired Bottom-Up Saliency Models in Vision Transformers

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Abstract

Vision Transformers (ViTs) (Dosovitskiy et al., 2021) have achieved strong performance in visual recognition tasks, but their uniform treatment of input tokens in early layers leads to inefficiencies in representation learning. Inspired by biological vision systems, which prioritize visually salient regions through bottom-up attention, this work investigates the role of deterministic saliency as an external prior for guiding transformer behavior.

We integrate a frequency-domain saliency model (ImgSig) into Vision Transformers using four distinct strategies: external token gating, adaptive input-level blending, feature-level injection, and attention-level modulation. These approaches enable systematic exploration of how saliency influences representations, token interactions, and model performance at different stages of the transformer pipeline.

Experimental results on CIFAR-100 show that saliency can improve performance when applied appropriately, with adaptive input-level blending achieving the best accuracy of approximately 91.44%. We find that saliency is most effective in early layers, where representations align with low-level visual structure, while its influence diminishes in deeper layers due to a mismatch with semantic features. Furthermore, adaptive mechanisms that allow the model to selectively utilize saliency outperform fixed integration strategies.

Our findings demonstrate that bottom-up saliency acts as an efficiency-enforcing inductive bias for Vision Transformers, providing insights into how biologically-inspired priors can guide modern deep learning architectures.

1 Introduction

Vision Transformers (ViTs) (Dosovitskiy et al., 2021) have emerged as a powerful alternative to convolutional neural networks for visual recognition tasks, achieving state-of-the-art performance across a wide range of benchmarks. By modeling global dependencies through self-attention (Vaswani et al., 2017), ViTs are able to capture long-range interactions between image patches, enabling flexible and expressive representations. However, a fundamental limitation of standard ViTs is their tendency to treat all input tokens uniformly during early stages of processing, leading to inefficiencies in both computation and representation learning.

In contrast, biological vision systems (Itti et al., 1998) exhibit highly efficient processing mechanisms by selectively allocating attention to visually salient regions of a scene. Bottom-up visual saliency, driven by low-level features such as edges, contrast, and texture, allows the human visual system to rapidly prioritize important regions without requiring task-specific knowledge. This mechanism serves as an efficient inductive bias, guiding perception toward informative structures while suppressing irrelevant background information.

Motivated by these observations, this work investigates whether biologically-inspired bottom-up saliency can serve as an external prior to improve the efficiency and behavior of Vision Transformers. Specifically, we explore the integration of a deterministic, class-agnostic saliency model (ImgSig) into the transformer pipeline, and analyze its impact on representation learning, attention dynamics, and overall model performance.

A key challenge in incorporating saliency into transformer architectures lies in determining *where* and *how* the saliency signal should be introduced. Unlike convolutional architectures, where spatial locality is inherently preserved, transformers operate on tokenized representations with flexible interaction patterns. As a result, saliency can influence the model at multiple levels, including input representations, intermediate feature transformations, and attention mechanisms.

To systematically study this design space, we propose and evaluate multiple saliency integration strategies. These include (i) external token gating, where saliency modulates token magnitudes prior to attention computation, (ii) adaptive input blending using learnable parameters that allow the model to control its reliance on saliency, (iii) layer-wise feature injection that incorporates saliency into intermediate representations, and (iv) attention-level modulation, where saliency directly biases the attention distribution.

Through extensive experiments on image classification tasks, we analyze the effectiveness of these approaches and identify key patterns governing the interaction between saliency and transformer representations. Our findings reveal that saliency is most beneficial when applied in early stages of the model, where representations remain closely aligned with low-level visual structure. Furthermore, adaptive mechanisms that allow the model to selectively incorporate saliency outperform fixed integration strategies, highlighting the importance of context-dependent saliency utilization.

The main contributions of this work are as follows:

- We introduce a systematic framework for integrating bottom-up saliency into Vision Transformers across multiple architectural levels, including input, feature, and attention spaces.
- We propose and evaluate a family of saliency-guided models, including adaptive blending mechanisms that learn when to trust saliency signals.
- We provide empirical evidence that bottom-up saliency improves representation learning primarily in early transformer layers, and characterize its limitations in deeper semantic stages.
- We offer a unified interpretation of saliency as an efficiency-enforcing prior, bridging insights from biological vision and modern transformer architectures.

2 Related Work

2.1 Vision Transformers

Vision Transformers (ViTs) were introduced by Dosovitskiy et al. (Dosovitskiy et al., 2021), extending the transformer architecture of Vaswani et al. (Vaswani et al., 2017) to image recognition by representing images as sequences of patches. Unlike convolutional neural networks (CNNs), which rely on inductive biases such as locality and translation invariance, ViTs process images as sequences of tokens and learn relationships through attention mechanisms. While this enables flexible and expressive representations, it also introduces challenges, particularly in terms of computational efficiency and the lack of inherent spatial bias in early layers.

Several works have explored improving the efficiency of ViTs by reducing redundant computation or focusing attention on important tokens. Methods such as token pruning, token merging, and dynamic attention mechanisms, including DynamicViT (Rao et al., 2021), aim to selectively process informative regions while discarding less relevant ones. These approaches highlight the need for mechanisms that can guide attention toward salient regions in an efficient and principled manner.

2.2 Saliency Models in Computer Vision

Visual saliency has been extensively studied in both neuroscience and computer vision as a mechanism for identifying regions of interest in an image. Early biologically-inspired models, such as the work of Itti et al. (Itti et al., 1998), detect salient regions using low-level cues like contrast, orientation, and spatial discontinuities. Frequency-domain approaches have also been proposed; for example, Hou and Zhang (Hou and Zhang, 2007) introduced the spectral residual method, which detects salient regions by analyzing irregularities in the frequency spectrum. These models are deterministic and class-agnostic, making them suitable as general-purpose priors.

More recent approaches to saliency prediction leverage deep learning to produce task-specific saliency maps. These methods often rely on supervised training and are optimized for specific objectives, such as predicting human gaze or highlighting class-relevant regions. While effective, such top-down saliency models introduce dependencies on training data and model parameters, making them less suitable as independent priors for guiding model behavior.

2.3 Saliency and Attention Mechanisms

The relationship between saliency and attention has been explored in various deep learning architectures. In convolutional networks, attention mechanisms such as spatial attention and channel attention modules selectively emphasize informative features. In transformer-based models, attention is explicitly computed through query-key interactions, making it a natural point of integration for saliency signals.

Some works have attempted to incorporate saliency into attention mechanisms by modifying attention weights or introducing additional bias terms. However, these approaches often tightly couple the saliency signal with the model’s learned representations, making it difficult to disentangle the contribution of saliency from that of the underlying architecture. This limits interpretability and complicates analysis of the role of saliency.

2.4 Efficiency in Transformer Architectures

Improving efficiency in transformer models has been an active area of research. Techniques such as sparse attention, token pruning, and dynamic computation aim to reduce the quadratic complexity of attention mechanisms. Methods such as DynamicViT (Rao et al., 2021) introduce learnable modules to selectively prune tokens during inference, effectively focusing computation on important regions.

These approaches implicitly rely on the assumption that not all tokens are equally important. However, they typically learn importance from data, rather than leveraging external priors. This raises the question of whether biologically-inspired saliency signals can serve as a complementary mechanism for guiding efficient computation in transformers.

2.5 Positioning of This Work

In contrast to prior work, this study investigates the use of bottom-up saliency as an external, deterministic prior for guiding Vision Transformers. Rather than learning saliency jointly with the model, we integrate a fixed saliency signal (ImgSig) at different stages of the transformer pipeline and systematically analyze its effect on representation learning and attention behavior.

Our approach differs from existing methods in three key ways. First, we focus on bottom-up saliency, which is independent of task-specific supervision and remains constant throughout training. Second, we explore multiple integration strategies, including input-level blending, feature-level injection, and attention-level modulation, providing a comprehensive view of the design space. Third, we emphasize interpretability and analysis, aiming to understand *where* and *how* saliency contributes to transformer performance, rather than solely optimizing accuracy.

3 Biological and Computational Saliency

3.1 Bottom-Up and Top-Down Attention

Visual attention in biological systems is typically described as the result of two complementary processes: bottom-up and top-down attention. Bottom-up attention is driven by stimulus properties such as contrast, edges, and texture, and operates independently of task-specific goals, as formalized in classic saliency-based models of visual attention (Itti et al., 1998). It enables rapid identification of visually distinctive regions in a scene and is often associated with early stages of visual processing. In contrast, top-down attention is guided by task demands, prior knowledge, and semantic understanding, allowing the observer to focus on regions relevant to a specific objective.

In the context of deep learning, top-down attention is commonly realized through learned attention mechanisms, where importance is inferred from data during training. Bottom-up saliency, however, remains underexplored as an external prior that can guide model behavior independently of learned representations. This distinction is particularly relevant when integrating saliency into pretrained transformer architectures, where stability and interpretability are critical considerations.

3.2 The ImgSig Saliency Model

To model bottom-up saliency, we employ the ImgSig method, a frequency-domain approach that captures structural information in images using simple and deterministic transformations, conceptually related to spectral saliency methods such as the spectral residual approach of Hou and Zhang (Hou and Zhang, 2007). Given an input image $x \in \mathbb{R}^{B \times C \times H \times W}$, where B is the batch size, C is the number of image channels, H is the image height, and W is the image width, the saliency map is computed as follows:

$$x \xrightarrow{\text{DCT}} X \xrightarrow{\text{sign}} \text{sign}(X) \xrightarrow{\text{iDCT}} \tilde{x} \quad (1)$$

The discrete cosine transform (DCT) converts the image into the frequency domain, where the sign operation retains only structural information while discarding magnitude. The inverse DCT then reconstructs an image that emphasizes edges and high-frequency components. To obtain a saliency map, we compute the local energy using element-wise multiplication, denoted by $\odot$:

$$e = \tilde{x} \odot \tilde{x} \quad (2)$$

This energy map is smoothed using a Gaussian filter and aggregated across channels to produce a single-channel saliency map $S \in \mathbb{R}^{B \times 1 \times H \times W}$. Finally, per-image min-max normalization is applied to map saliency values to the range $[0, 1]$.

3.3 Properties of Bottom-Up Saliency

The ImgSig saliency model exhibits several properties that make it suitable as an external prior for Vision Transformers.

Class-agnostic behavior. Unlike supervised saliency methods, ImgSig does not rely on labels or task-specific objectives. As a result, it provides a general-purpose measure of visual importance that is independent of the downstream task.

Focus on low-level structure. The use of frequency-domain transformations emphasizes edges, textures, and spatial discontinuities. These features align closely with early-stage visual processing and the initial layers of transformer models.

Deterministic and stable. Since ImgSig does not involve learned parameters, the saliency signal remains fixed throughout training. This avoids feedback loops and ensures that observed performance changes can be attributed to the integration strategy rather than changes in the saliency model itself.

Computational simplicity. The saliency computation involves only basic linear transforms and filtering operations, making it efficient and easy to integrate into existing pipelines without significant overhead.

3.4 From Saliency Maps to Token-Level Signals

Vision Transformers operate on tokenized representations obtained by partitioning the input image into patches. To integrate saliency into this framework, the spatial saliency map must be converted into a token-level signal.

Given a saliency map $S \in \mathbb{R}^{H \times W}$ and a patch size (p_h, p_w) , we compute a per-token saliency value by average pooling over each patch region. We refer to this operation as *patch pooling* in the architecture diagrams. Patch pooling converts the pixel-level saliency map into token-level saliency values by averaging all saliency scores inside the spatial region corresponding to each ViT patch:

$$s_i = \frac{1}{p_h p_w} \sum_{(u,v) \in \text{patch}_i} S(u, v) \quad (3)$$

This results in a vector $s \in [0, 1]^N$, where N is the number of tokens. Each token is thus associated with a scalar saliency value indicating its relative visual importance.

3.5 Role of Saliency as an External Prior

In this work, saliency is treated as an external prior rather than a learned component of the model. This design choice is motivated by two considerations. First, it allows us to study the effect of saliency independently of model training, enabling controlled experiments across different integration strategies. Second, it ensures compatibility with pretrained Vision Transformers, as the saliency signal does not interfere with learned attention patterns during early training stages.

By maintaining a clear separation between the saliency signal and the model parameters, we are able to systematically investigate how and where saliency contributes to representation learning and attention dynamics within transformer architectures.

4 Saliency Integration Strategies in Vision Transformers

4.1 Overview of Design Space

A key challenge in incorporating saliency into Vision Transformers is determining how and where the saliency signal should be integrated into the model. Unlike convolutional networks, where spatial locality provides a natural mechanism for integrating saliency, transformer architectures operate on tokenized representations with flexible interaction patterns.

We identify three primary levels at which saliency can influence a transformer: (i) the input representation, where saliency modifies token embeddings before attention is computed, (ii) the feature representation, where saliency is injected into intermediate layers, and (iii) the attention mechanism itself, where saliency directly biases token interactions. In addition, we consider adaptive mechanisms that allow the model to learn how much to rely on saliency signals.

To systematically explore this design space, we evaluate four integration strategies: external gating (Case-A), adaptive input blending (Case-1), feature-level injection (Case-2), and attention-level modulation (Case-3).

4.2 Case-A: External Saliency Gating

In the simplest integration strategy, saliency is used as an external gating signal applied to token embeddings prior to attention computation. Given token representations $X \in \mathbb{R}^{N \times D}$ and token-level saliency values $s \in [0, 1]^N$, where N is the number of patch tokens and D is the token embedding dimension, we define a gating function:

$$g_i = (1 - w) + w \cdot s_i^\alpha \quad (4)$$

where $w \in [0, 1]$ is a gating weight that is gradually increased during training, and α controls the sharpness of the saliency distribution.

The gated tokens are then computed as:

$$\tilde{X} = \text{LN}(X) \odot g \quad (5)$$

where $\text{LN}(\cdot)$ denotes Layer Normalization and $\odot$ represents element-wise multiplication. Layer Normalization normalizes each token across its feature dimension D before scaling, which helps stabilize the magnitude of token activations before saliency gating is applied. Since $g \in \mathbb{R}^N$ contains one scalar per token, it is broadcast across the embedding dimension when multiplied with $\text{LN}(X) \in \mathbb{R}^{N \times D}$.

This approach preserves the original attention mechanism while modulating the magnitude of token representations. We evaluate different placement strategies and observe that applying gating only in early layers yields the best performance, suggesting that saliency aligns primarily with low-level features.

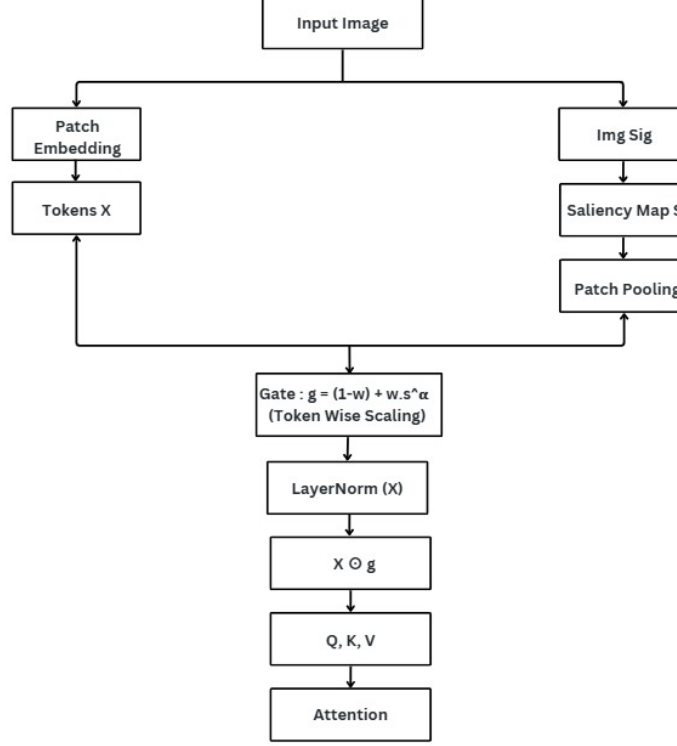


Figure 1: Case-A: External saliency gating applied to token embeddings before attention computation.

4.3 Case-1: Adaptive Input-Level Blending

To overcome the rigidity of fixed gating, we introduce an adaptive blending mechanism that allows the model to learn how much to rely on saliency. Instead of directly scaling tokens, we construct a blended representation:

$$X' = (1 - \beta)X + \beta S \quad (6)$$

where S denotes the saliency-aligned token representation and $\beta \in [0, 1]$ is a learnable parameter controlling the contribution of saliency. This differs from Case-A in Equation (4): Case-A performs multiplicative gating, which scales the magnitude of existing token features, whereas Case-1 performs additive interpolation between the original token representation X and the saliency-aligned representation S . Thus, Case-A asks how strongly each token should be scaled, while Case-1 asks how much of the representation should come from the saliency stream.

We explore several parameterizations of β :

- **Level 0:** Global scalar β shared across all tokens and images.
- **Level 1:** Token-wise static β_i , allowing spatial variation.
- **Level 2:** Token-wise, image-dependent $\beta_i = \sigma(\text{MLP}(x_i))$, enabling adaptive control based on feature context. Here, MLP denotes a multi-layer perceptron, a small feedforward neural network that maps each token feature x_i to a scalar blending weight.

- **Level 3:** Image-level β , dependent on global image features.
- **Level 4:** Attention-derived β , computed from attention statistics, such as normalized class-token-to-patch attention scores from an early transformer block. This tests whether the model’s own attention patterns can act as a proxy for saliency trust.
- **Level 5:** Agreement-based $\beta = f(X, S)$, incorporating both feature and saliency information. In this case, $f(\cdot)$ is a lightweight learnable function that receives both the image token representation and saliency-aligned representation and predicts whether the two sources agree for each token.

Among these variants, the token-wise, image-dependent formulation (Level 2) achieves the best performance, highlighting the importance of context-aware saliency integration.

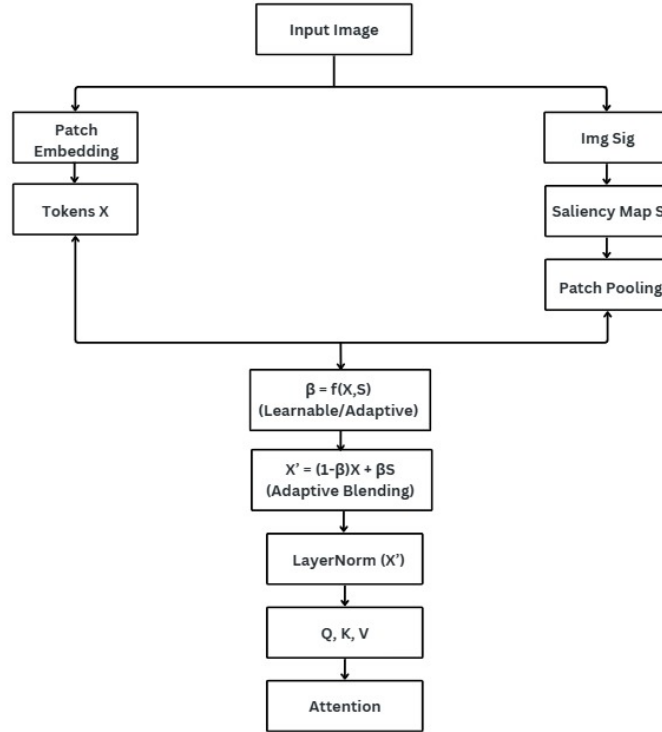


Figure 2: Case-1: Adaptive input-level blending using a learnable parameter β to control saliency contribution. Patch pooling denotes averaging the dense saliency map within each ViT patch to obtain token-level saliency. Arrows indicate information flow from the input image through the image-token and saliency-token branches before merging at the adaptive blending block.

4.4 Case-2: Feature-Level Injection

In this strategy, saliency is injected directly into intermediate feature representations within the transformer. For a given layer l , the feature representation is modified as:

$$\tilde{X}^{(l)} = X^{(l)} + \alpha (X^{(l)} \odot S) \quad (7)$$

where α controls the strength of the saliency contribution.

This formulation can be interpreted as a residual bias that reinforces salient regions while preserving the original representation. We evaluate different layer-wise injection schemes and find that saliency improves performance when applied to early layers, but degrades performance when extended to deeper layers. This suggests a mismatch between low-level saliency signals and high-level semantic features.

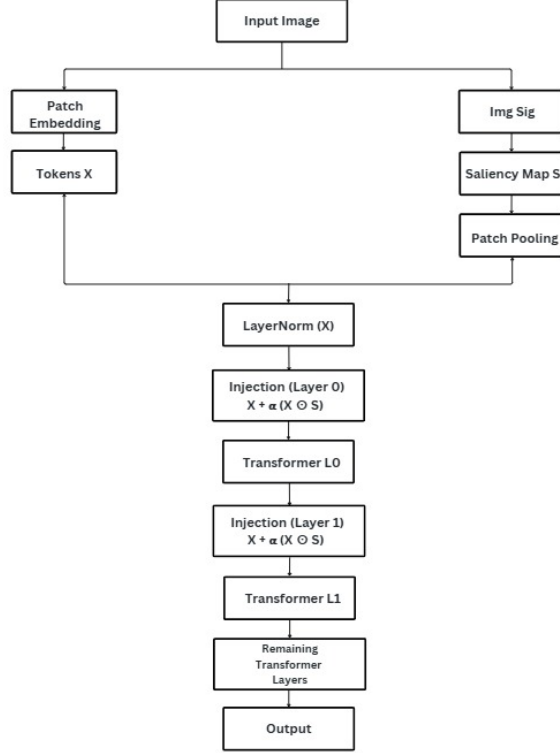


Figure 3: Case-2: Layer-wise feature-level saliency injection into intermediate transformer representations. Arrows indicate top-to-bottom processing, with the saliency branch producing patch-level saliency values that are injected into selected transformer layers.

4.5 Case-3: Attention-Level Modulation

In the final strategy, saliency is integrated directly into the attention mechanism by modifying the attention logits in the standard scaled dot-product attention framework introduced by Vaswani et al. (Vaswani et al., 2017). In this context, attention logits refer to the raw pre-softmax token compatibility scores, $QK^T/\sqrt{d}$; they are not log-odds, but unnormalized similarity values that are converted into attention probabilities by the softmax function. Given queries Q and keys K , the attention weights are computed as:

$$A = \text{softmax} \left(\frac{QK^T}{\sqrt{d}} + \alpha\Phi(S) \right) \quad (8)$$

where d is the dimensionality of each attention head and $\Phi(S)$ is a transformation that maps

token-level saliency values to a pairwise bias term compatible with the attention logits. The softmax is applied row-wise; for a vector z , $\text{softmax}(z_i) = \exp(z_i) / \sum_j \exp(z_j)$, producing a probability distribution over keys for each query token. In our implementation, a simple choice for the saliency bias is $\Phi(S) = SS^\top$. If $S \in \mathbb{R}^{N \times 1}$ contains one saliency value per token, then $SS^\top \in \mathbb{R}^{N \times N}$ and each entry $(SS^\top)_{ij} = s_i s_j$ increases the attention compatibility between pairs of salient tokens.

This approach directly influences how tokens attend to one another, rather than modifying the token representations themselves. While this method leads to consistent utilization of saliency across layers, its performance is limited by the quality and semantic alignment of the saliency signal.

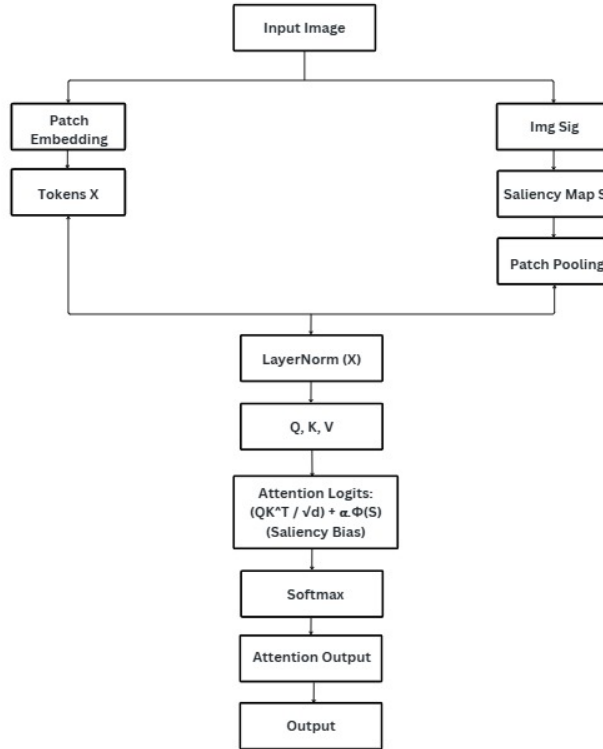


Figure 4: Case-3: Attention-level saliency modulation by adding a bias term to attention logits. Arrows indicate that image tokens produce Q , K , and V , while patch-pooled saliency produces a pairwise saliency bias that is added before the row-wise softmax.

Table 1: Comparison of saliency integration strategies across architectural levels and effects.

Case	Level	Adaptivity	Target	Primary Effect
Case-A	Pre-attention	Fixed	Tokens	Magnitude scaling
Case-1	Input	Adaptive (β)	Features	Context-aware fusion
Case-2	Intermediate	Fixed (α)	Features	Representation bias
Case-3	Attention	Fixed (α)	Attention	Interaction modulation

4.6 Unified Interpretation

The four integration strategies explored in this work can be understood as influencing different components of the transformer pipeline:

- **Case-A:** Modifies token magnitudes prior to attention.
- **Case-1:** Blends saliency with token representations at the input level.
- **Case-2:** Injects saliency into intermediate feature representations.
- **Case-3:** Biases the attention mechanism directly.

These approaches collectively illustrate that saliency can act at multiple levels within a transformer, affecting both representation and interaction. Our analysis reveals that the effectiveness of saliency depends critically on the stage of integration, with early-layer and adaptive methods providing the most consistent improvements.

5 Experimental Setup and Results

5.1 Dataset and Task

We evaluate our methods on the CIFAR-100 dataset (?), which consists of 60,000 color images of size 32×32 across 100 classes, with 50,000 training images and 10,000 test images. Despite its relatively small size, CIFAR-100 provides a challenging multi-class classification task that is useful for controlled comparison of saliency integration strategies. However, CIFAR-100 is not an ideal benchmark for fully evaluating attention selection, because its images are small and usually contain relatively simple scenes with limited object complexity. As a result, there may be less need for sophisticated attentional selection than in larger datasets such as ImageNet or datasets with multiple objects per image. We therefore treat CIFAR-100 as a controlled proof-of-concept setting rather than a complete evaluation of saliency-guided attention.

5.2 Model Architecture

We use a standard Vision Transformer (ViT) as the backbone model. The input images are partitioned into non-overlapping patches and embedded into token representations, which are then processed through a sequence of transformer encoder blocks consisting of multi-head self-attention and feedforward layers. A classification token is used for final prediction.

All saliency integration strategies described in Section 4 are implemented on top of this baseline architecture without modifying the core transformer design.

5.3 Training Setup

All models are trained using the AdamW optimizer (?) with separate learning rates for the backbone and classification head. The backbone is initialized with pretrained weights, and a lower

learning rate is used to preserve learned representations, while the classification head is trained with a higher learning rate.

We employ standard data augmentation techniques, including random cropping and horizontal flipping. Additionally, we use Mixup (?) and CutMix (?) to improve generalization. Mixed precision training (?) is used to improve computational efficiency, and exponential moving average (EMA) of model weights, related to Polyak averaging (?), is maintained to stabilize evaluation.

For Case-A, a gating warm-up schedule is used, where the saliency influence is gradually increased during the initial epochs to prevent disruption of pretrained features.

5.4 Evaluation Metrics

We report classification accuracy as the primary performance metric. In addition, we analyze attention behavior qualitatively and quantitatively to understand how saliency affects token importance and feature learning. For selected experiments, we also examine metrics such as area under the curve (AUC) to assess robustness.

5.5 Results: Case-A (External Gating)

We first evaluate the external saliency gating approach. Applying saliency gating only in the first transformer layer yields the best performance, achieving approximately 90.36% accuracy. Extending gating to deeper layers results in a performance drop, with full-layer gating achieving approximately 89.6% accuracy. Randomized gating shows marginal improvements at low probabilities but degrades performance when applied excessively.

These results indicate that bottom-up saliency aligns most effectively with early-stage representations.

5.6 Results: Case-1 (Adaptive Blending)

Adaptive input-level blending significantly improves performance compared to fixed gating. Among the different parameterizations of β , the token-wise, image-dependent formulation (Level 2) achieves the highest accuracy of approximately 91.44%.

Simpler variants, such as global or token-wise static β , show moderate improvements, while more complex formulations, including attention-derived and agreement-based β , do not provide additional gains. This suggests that adaptive, context-dependent control of saliency is critical for effective integration.

5.7 Results: Case-2 (Feature-Level Injection)

Feature-level saliency injection demonstrates that the effectiveness of saliency is highly dependent on depth. Injecting saliency into the first layer yields an accuracy of approximately 89.30%, while injecting into the first two layers improves performance to approximately 89.55%. However, extending injection to deeper layers leads to performance degradation.

These results highlight a mismatch between low-level saliency signals and higher-level semantic representations.

5.8 Results: Case-3 (Attention-Level Modulation)

Attention-level saliency modulation results in consistent but modest improvements. Injecting saliency into the first layer achieves approximately 89.30% accuracy, while extending to the first two layers improves performance slightly to approximately 89.36%.

Interestingly, this approach demonstrates that the model is capable of utilizing saliency across layers, although the overall performance gain is limited by the quality of the saliency signal.

5.9 Comparative Analysis

Across all methods, adaptive input-level blending (Case-1) achieves the best performance, followed by external gating (Case-A). Feature-level and attention-level integration strategies provide insights into how saliency interacts with internal representations but do not outperform adaptive blending.

A consistent pattern across all experiments is that saliency is most beneficial when applied in early layers. As representations become more semantic in deeper layers, the usefulness of bottom-up saliency diminishes, leading to reduced performance when applied excessively.

5.10 Summary of Findings

The experimental results demonstrate that bottom-up saliency can improve Vision Transformer performance when integrated appropriately. However, its effectiveness is highly dependent on both the location and method of integration. Adaptive mechanisms that allow the model to selectively incorporate saliency signals provide the most robust improvements, while rigid or overly deep integration strategies can be detrimental.

Table 2: Comprehensive evaluation of saliency integration strategies on CIFAR-100.

Method	Configuration	Accuracy (%)
Baseline	ViT (no saliency)	86.90
Case-A (Gating)	Layer 0 only	90.36
	All layers	89.60
	Random layers (low probability)	~89.70
Case-1 (Adaptive Blending)	Level 0 (Scalar β)	88.75
	Level 1 (Token-wise static β)	89.10
	Level 2 (Token-wise MLP β)	91.44
	Level 3 (Image-level β)	90.44
	Level 4 (Attention-derived β)	89.57
	Level 5 (Agreement-based β)	89.17
Case-2 (Feature Injection)	Layer 0	89.30
	Layers (0,1)	89.55
	Layers (0,1,2)	88.90
Case-3 (Attention Modulation)	Layer 0	89.30
	Layers (0,1)	89.36

As shown in Table 2, adaptive blending (Case-1) achieves the best performance. Compared with the baseline ViT accuracy of 86.90%, all saliency-based methods improve performance when applied appropriately. Case-A layer-0 gating improves the baseline by 3.46 percentage points, while the best Case-1 model improves it by 4.54 percentage points. Case-2 and Case-3 provide smaller but still positive gains, with their best settings improving the baseline by 2.65 and 2.46 percentage points, respectively. These comparisons show that the benefit of saliency is not simply due to adding extra parameters, but depends strongly on where and how the saliency prior is integrated.

6 Discussion and Key Insights

6.1 Saliency as an Efficiency-Enforcing Prior

A central motivation of this work is to investigate whether bottom-up saliency can act as an efficiency-enforcing prior in Vision Transformers. Our results suggest that saliency provides a meaningful inductive bias that encourages the model to focus on informative regions of the input. By modulating token representations or attention patterns, saliency effectively reduces the influence of less relevant regions, implicitly guiding the model toward more efficient use of its representational capacity.

However, unlike explicit pruning methods, saliency does not directly reduce computational complexity in our current formulation. Instead, it shapes the distribution of attention and feature activations, indicating that saliency may serve as a precursor to more aggressive efficiency mechanisms such as token pruning or dynamic computation, as explored in approaches like DynamicViT (Rao et al., 2021).

6.2 Depth-Dependent Effectiveness of Saliency

A consistent observation across all experiments is that the effectiveness of saliency is strongly dependent on the depth at which it is applied. Early layers benefit the most from saliency integration, while deeper layers often experience performance degradation when saliency is applied.

This behavior can be explained by the nature of representations in transformer models. Early layers primarily encode low-level visual features such as edges and textures, which are well-aligned with bottom-up saliency signals. In contrast, deeper layers capture higher-level semantic information, where the relevance of regions depends on class-specific context rather than purely structural cues. As a result, bottom-up saliency becomes increasingly misaligned with the representations in deeper layers.

6.3 Adaptive vs Fixed Saliency Integration

Another key finding is the importance of adaptivity in saliency integration. Fixed strategies, such as uniform gating or repeated feature injection, assume that saliency is uniformly useful across tokens and images. However, our results demonstrate that this assumption does not hold in practice.

Adaptive mechanisms, particularly the token-wise, image-dependent blending approach in Case-1, allow the model to selectively incorporate saliency based on context. This flexibility enables the model to use saliency when it is informative and ignore it when it is not, leading to

improved performance and stability. In contrast, rigid integration strategies often overemphasize saliency, resulting in information loss or interference with learned representations.

6.4 Representation-Level vs Attention-Level Integration

Our experiments also reveal important differences between representation-level and attention-level integration of saliency. Feature-level injection (Case-2) modifies the internal representations of the model, while attention-level modulation (Case-3) directly influences how tokens interact with one another.

Although attention-level integration is conceptually aligned with the role of saliency as an importance signal, its effectiveness is limited by the quality of the saliency map. Since bottom-up saliency lacks semantic awareness, its influence on attention may not always correspond to task-relevant relationships between tokens. In contrast, representation-level approaches can more naturally incorporate structural biases without directly interfering with attention computations.

6.5 Limitations of Bottom-Up Saliency

While bottom-up saliency provides a useful prior, it has inherent limitations. Most notably, it is class-agnostic and does not incorporate semantic information about the task. As a result, it may highlight regions that are visually distinctive but not relevant for classification. Another limitation is the use of CIFAR-100 as the main evaluation dataset. Because CIFAR-100 images are small and typically contain limited scene complexity, the dataset may not fully stress-test attentional selection mechanisms. Larger datasets with higher-resolution images and more complex scenes would provide a stronger setting for evaluating whether saliency improves spatial selection and attention allocation.

This limitation becomes particularly pronounced in deeper layers, where semantic understanding is critical. The mismatch between low-level saliency and high-level features suggests that bottom-up saliency alone is insufficient for guiding attention in later stages of the model.

6.6 Implications for Hybrid Attention Mechanisms

The observed limitations of bottom-up saliency suggest a promising direction for future work: combining bottom-up and top-down attention mechanisms. Bottom-up saliency can provide an initial structural prior in early layers, while top-down signals derived from the model or task can guide attention in deeper layers.

Such hybrid approaches may enable more effective and efficient transformer architectures by leveraging the strengths of both types of attention. In particular, integrating learned, semantic saliency with fixed structural saliency could provide a more comprehensive representation of visual importance.

6.7 Summary of Insights

The key insights from this work can be summarized as follows:

- Bottom-up saliency acts as a useful inductive bias that improves representation learning when applied appropriately.
- The effectiveness of saliency is strongly depth-dependent, with early layers benefiting the most.
- Adaptive integration mechanisms outperform fixed strategies by allowing context-dependent use of saliency.
- Representation-level and attention-level integration have distinct advantages and limitations.
- Bottom-up saliency alone is insufficient for capturing semantic relevance, motivating hybrid approaches.

7 Conclusion

In this work, we investigated the role of biologically-inspired bottom-up saliency as an external prior for Vision Transformers. By integrating a deterministic saliency model (ImgSig) into different components of the transformer architecture, we systematically explored how saliency influences representation learning, attention dynamics, and overall model performance.

Our experiments demonstrate that bottom-up saliency can improve performance when incorporated appropriately, particularly in the early stages of the model where representations remain aligned with low-level visual structure. Among the proposed methods, adaptive input-level blending consistently achieved the best results, highlighting the importance of allowing the model to selectively utilize saliency signals rather than enforcing them uniformly.

We further showed that the effectiveness of saliency is strongly dependent on the level of integration. While input-level and early feature-level integration provide meaningful improvements, extending saliency into deeper layers leads to performance degradation due to the mismatch between low-level saliency cues and high-level semantic representations. Additionally, attention-level modulation, although conceptually appealing, is limited by the lack of semantic information in bottom-up saliency.

These findings suggest that bottom-up saliency acts as an efficiency-enforcing inductive bias that guides models toward more informative regions of the input without requiring additional supervision. However, its limitations also indicate that purely bottom-up approaches are insufficient for capturing task-specific relevance in deeper layers.

Overall, this work provides a unified framework for understanding how saliency interacts with transformer architectures and offers practical insights into where and how saliency should be integrated. We believe that these findings can inform the design of more efficient and interpretable vision models, particularly when combined with complementary mechanisms such as top-down attention or dynamic token selection.

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